

Purpose of Homoscedasticity and Heteroscedasticity

Homoscedasticity:

Homoscedasticity means the error (residuals) of your regression model has constant variance across all predicted values as the spread of errors being uniform throughout.

Heteroscedasticity:

Heteroscedasticity means the error (residuals) has non-constant variance as the spread of errors increases or decreases as the predicted values change.

Purpose:

The purpose is to validate one of the core assumptions of linear regression that residuals are equally spread. If this assumption is violated (i.e., heteroscedasticity exists), your model's coefficient estimates become unreliable and hypothesis tests give wrong results.

When do we need to check it:

- After building a linear regression model, before trusting its results.
- When your residual plot shows a funnel shape (wider on one end) that is a sign of heteroscedasticity.
- When working with financial data, salary data, or any data where values vary widely across ranges.

NOTES:

Homoscedasticity = Good and the model is healthy

Heteroscedasticity = Bad and the model needs a fix